2025 PRIME ATM Winter

Paly Math Club

November 2025

Instructions

- Welcome to our Annual Team Math tournament, drafted for winter!
- This is a **40** minute test with **21** questions, the last question being one that tests your estimation skills.
- Please do not begin until instructed to do so.
- No materials other than simple writing utensils may be permitted.
- Each question is worth one point.
- The questions **may not be in difficulty order**.
- There are no point deductions for incorrect answers.
- There are no partial credit except the estimation problem.
- Please **box** your answers in the spaces below the corresponding question.
- Scores will be released to all who participated in a few days **via email**.

Team Name: (0.5pt) _____

Team Member Names: (0.5pt) _____

1 Problems

1. If $4x - 28 = -24$, what is the value of $x - 7$?

2. The function $f(x)$ is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

3. The two sides of an isosceles triangle $\triangle ABC$ have lengths 9 and 4. What is the area of $\triangle ABC$?

4. The arithmetic mean of numbers 2, 6, 8, and x is 7. The arithmetic mean of numbers 2, 6, 8, x, and y is 9. What is $y - x$?

5. A line l is drawn through a square such that it splits the square into two regions, each with area 32. If the line intersects the square 1 unit from one vertex, let A be the length of the portion of l contained entirely within the square. Find A.

6. When randomly arranging the letters in PALOALTO, what is the probability that no two identical letters are adjacent to each other?

7. Aiden owns 6 cards, each labeled with numbers from 1 to 6. He places these cards on a single row in arbitrary order. What is the probability that the sum of the numbers on the first and last cards is greater than 10?

8. Suppose that there are 10 people, labelled from 1 to 10. The first 9 people each claim the next person is lying (e.g Person 1 claims Person 2 is lying, Person 2 claims Person 3 is lying, so on). The last person claims that no one is lying. What is the product of the labels on the people that tell the truth?

9. Use the symbols $+, -, \times, \div$ and parentheses to create 24 using the numbers $3, 3, 7, 7$. (Write your solution below)

10. In rhombus ABCD, $\angle BAD = 60°$. When we folded $\triangle ABD$ across BD to create a tetrahedron, the altitude that extended from D to the plane ABC landed on the midpoint of AC. If the resulting angle between the planes ABD and CBD is θ, find $\cos(\theta)$.

11. Let v, w, x, y, and z be integers such that $2^v + 2^w + 2^x + 2^y + 2^z = 280.375$. Find the product $vwxyz$.

12. Suppose we write the odd positive integers in a line from left to right in increasing order (13579111315...). What is the 2025th digit in the line?

13. 2^{29} is a 9-digit number with all distinct digits. Which of the 10 decimal digits does not appear in it?

14. Each of the three people Alice, Bob, and Charlie each have a 2/3 chance of telling the truth each time they speak. Suppose Alice says something and Bob and Charlie both say she is lying. What is the chance Alice actually had told the truth?

15. A truncated icosidodecahedron is a convex 3D solid with 62 faces (30 squares, 20 regular hexagons, and 12 regular decagons, all with equal side length). What is the sum of the number of edges and the number of vertices of this solid?

16. Let θ be a real number such that $1 + \sin(2\theta) - (\frac{1}{2}\sin(2\theta))^2 = 0$. Compute the maximum value of $(1 + \sin\theta)(1 + \cos\theta)$.

17. Six people sit around a round table. Each person rolls a standard 6-sided die. If no two people sitting next to each other rolled the same number, we will say that the roll is valid. How many different rolls are valid?

18. Let ABCD be a regular tetrahedron and let W, X, Y, Z denote the centers of faces ABC, BCD, CDA, and DAB, respectively. What is the ratio of the volumes of tetrahedrons W XY Z and W AY Z? Express your answer as a decimal or a fraction in lowest terms.

19. Compute the sum of the positive integers $n \leq 100$ for which the polynomial $x^n + x + 1$ can be written as the product of at least 2 polynomials of positive degree with integer coefficients.

20. The area of right triangle ABC is 4, and hypotenuse AB is 12. Compute the perimeter of ABC.

ESTIMATION - TIEBREAKER

21. How many whole basketballs fit in the PALY Big Gym? (integer answers please)

SCRATCH PAPER (NOT GRADED UNLESS ASKED TO)